Class 10 NCERT Maths

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Chapter 13: Statistics

CHAPTER 13: STATISTICS

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 STATISTICS KYA HAI?

STATISTICS = data ko collect, organize, aur analyse karne ki vidya. Is chapter mein GROUPED data (intervals mein bante data) ka central tendency nikaalte hain.

1.2 TEEN CENTRAL TENDENCY MEASURES

MEAN (average): sab values ka total / count
MEDIAN: beech wali value (sort karne ke baad)
MODE: sabse zyada baar aane wali value

1.3 GROUPED DATA KE TERMS

Example data:

Class Interval	Frequency (fi)
0-10	**5**
10-20	**8**
20-30	**12**

- CLASS INTERVAL: range (jaise 0-10)
- FREQUENCY (fi): us interval mein kitne items
- CLASS MARK (xi): interval ka midpoint
  = (upper + lower)/2
  0-10 ka class mark = (0+10)/2 = 5

SECTION 2: MEAN OF GROUPED DATA (3 METHODS)

2.1 METHOD 1: DIRECT METHOD

Mean = Sum(fi xi) / Sum(fi)

STEPS:

1. Har interval ka class mark xi nikaalo
2. fi x xi nikaalo har row
3. Sum(fi xi) aur Sum(fi) ka total
4. Divide karo

EXAMPLE 1:

CI	fi	xi	fi*xi
0-10	**5**	**5**	**25**
10-20	**8**	**15**	**120**
20-30	**12**	**25**	**300**
30-40	**5**	**35**	**175**
Total	30		620

Mean = 620 / 30 = 20.67

2.2 METHOD 2: ASSUMED MEAN METHOD

Jab numbers bade ho, assumed mean 'a' le lo
(usually beech wala class mark), deviation di = xi - a.

Mean = a + (Sum(fi di) / Sum(fi))

2.3 METHOD 3: STEP DEVIATION METHOD

Jab class size h saman ho:

ui = (xi - a) / h

Mean = a + h x (Sum(fi ui) / Sum(fi))

(Calculation aur easy - chhote numbers)

SECTION 3: MODE OF GROUPED DATA

3.1 MODAL CLASS

MODAL CLASS = sabse zyada frequency wala interval.

3.2 FORMULA

Mode = l + [(f1 - f0) / (2 f1 - f0 - f2)] x h

l = modal class ka lower limit
f1 = modal class ki frequency
f0 = modal class se PEHLE wali frequency
f2 = modal class ke BAAD wali frequency
h = class size

3.3 EXAMPLE 2

CI	Frequency	
0-10	**3**	
10-20	8	<- f0
20-30	12	<- f1 (modal class)
30-40	7	<- f2

l = 20, f1 = 12, f0 = 8, f2 = 7, h = 10

Mode = 20 + [(12-8)/(2*12 - 8 - 7)] x 10
= 20 + [4/(24-15)] x 10
= 20 + (4/9) x 10
= 20 + 4.44
= 24.44

SECTION 4: MEDIAN OF GROUPED DATA

4.1 MEDIAN CLASS

Pehle CUMULATIVE FREQUENCY (cf) nikaalo (frequencies add karte jao). Phir n/2 dekho - jis class mein n/2 aata hai wahi MEDIAN CLASS.

4.2 FORMULA

Median = l + [(n/2 - cf) / f] x h

l = median class ka lower limit
n = total frequency (Sum fi)
cf = median class se PEHLE wali cumulative frequency
f = median class ki frequency
h = class size

4.3 EXAMPLE 3

CI	f	cf	
0-10	**5**	**5**	
10-20	**8**	**13**	
20-30	12	25	<- median class
30-40	**5**	**30**	

n = 30, so n/2 = 15
n/2 = 15 first crosses at cf = 25 -> class 20-30
l = 20, cf(before) = 13, f = 12, h = 10

Median = 20 + [(15 - 13)/12] x 10
= 20 + (2/12) x 10
= 20 + 1.67
= 21.67

SECTION 5: EMPIRICAL RELATIONSHIP

In teeno ke beech approximate relation:

3 Median = Mode + 2 Mean

(Ya: Mode = 3 Median - 2 Mean)

Ek nikaalna ho aur do pata ho toh ye use kar sakte ho.

Q AND A TIME

Q1. Class interval 30-40 ka class mark (xi) kya hoga?

Q2. Mean nikaal (direct method):
CI fi

0-10	**4**
10-20	**6**
20-30	**10**

Q3. Mode formula likho aur batao f0, f1, f2 kya hote.

Q4. Median formula mein cf ka kya matlab hai?

Q5. Agar Mean = 30 aur Median = 28, empirical relation se Mode nikaal.

Q6. Modal class aur median class mein kya difference?

SUMMARY

1. Class mark xi = (upper + lower)/2

2. Mean (Direct): xbar = Sum(fi xi) / Sum(fi)

3. Mean (Step dev): xbar = a + h(Sum(fi ui)/Sum fi), ui = (xi - a)/h

4. Mode: l + [(f1-f0)/(2 f1 - f0 - f2)] x h

5. Median: l + [(n/2 - cf)/f] x h

6. Empirical: 3 Median = Mode + 2 Mean

TIPS FOR EXAM

1. Table banao with columns: CI, fi, xi, fi*xi (mean), ya cf (median). Organized rehne se mistake nahi.

2. Mode: pehle modal class (max frequency) dhundo, phir formula. f0 aur f2 ko sahi pick karo.

3. Median: cumulative frequency table zaroor banao, n/2 cross karne wali class = median class.

4. Step deviation method bade numbers mein time bachata hai - prefer karo.

5. Empirical relation se shortcut nikaal sakte ho agar do measures diye ho.
